



Modelling Financial Markets

Seminar 2: Market Efficiency — Hands-on



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Acronyms used in this material (1/2)

- AI: Artificial Intelligence
- AMH: Adaptive Markets Hypothesis
- AR: AutoRegressive (model)
- BDS: Brock–Dechert–Scheinkman (test)
- BET: Bucharest Exchange Trading (index)
- BET-TR: BET Total Return (index)
- BIST: Borsa İstanbul (BIST 100 index) (the Istanbul Stock Exchange (BIST 100 index))
- BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CD: Chow–Denning (multiple variance-ratio test)
- CI: Confidence Interval
- COVID-19: COronaVirus Disease 2019
- CV: Cross-Validation
- DFA: Detrended Fluctuation Analysis
- FRED: Federal Reserve Economic Data
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- FWER: Family-Wise Error Rate
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- IPC: Índice de Precios y Cotizaciones (the Mexican Stock Exchange index)
- NW: Newey–West (standard errors)
- OLS: Ordinary Least Squares
- PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- RW1: Random Walk 1 (i.i.d. increments)
- RW3: Random Walk 3 (uncorrelated increments)
- SD: Standard Deviation

Acronyms used in this material (2/2)

- **SE: Standard Error**
- **TOM: Turn Of the Month**
- **US: United States**
- **VR: Variance Ratio**
- **WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)**

Today's Question and Required Outputs

- Question of the seminar: **which weak-form rejections of Chapter 2 survive robust inference, several horizons and several tests?**
- Route
 - ▶ A1, A5–A6: a variance ratio and an implied Hurst exponent on paper
 - ▶ B1, B7, C2: classical vs robust variance ratios, the crisis check, an AI answer to audit
 - ▶ B5, B6: a calendar effect with HAC errors, then fifteen tests with Holm
 - ▶ B3, B4: DFA and variance ratios on rolling windows with bootstrap bands
- Required outputs (checklist at the end of the core): the A worksheets, one table and one figure per B exercise, the C2 audit, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A1	what does $\widehat{VR}(2)$ compute, line by line?	Solved	–
A5, A6	does an implied $H(20) \neq 0.5$ prove long memory?	Solved, Proposed	A5
B1	do the S&P 500 and BET reject RW3 with robust errors?	Solved	–
B7	is the S&P 500 reversal a crisis effect?	Proposed	B1
C2	what is wrong in an AI variance-ratio answer?	Proposed	A1, B1
B5	is the turn-of-the-month effect real?	Solved	–
B6	which of 15 calendar effects survive Holm?	Proposed	B5
B3	does the S&P 500 DFA exponent leave its null band?	Solved	–
B4	has Bitcoin become more efficient over time?	Proposed	B3, B1
Extensions	A2, A3, A4, B2, B8, B9, C1, C3 (optional, after the seminar)	both	on each task

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: computation and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

Data Manifest

Series	Symbol, field	Calendar	Used in
S&P 500	GSPC.INDX, close	US trading days	A6, B1, B3, B5–B7, C2
BET	BET, close	BVB trading days	A5, B1, B2, B5, B6, C1
BET-TR and 8 emerging indices	BETTR.INDX, WIG20.INDX, BUX.INDX, PX.INDX, XU100.INDX, BVSP.INDX, MXX.INDX, NSEI.INDX, SSEC.INDX; close	own calendar each	B2, B6, C1
Euro Stoxx 50	STOXX50E.INDX, close	own calendar	B6
Bitcoin	BTC-USD.CC, close	every day	B4
Banca Transilvania, BRD	TLV.R0, BRD.R0, adjusted close	joined with BET	B9
S&P 500 monthly	Shiller S&P Composite, GSPC.INDX, SPY dividends; FRED TB3MS	month ends	B8
49 industry portfolios	Kenneth French Data Library , value-weighted, monthly	month ends	C3

- Market files data/market/<SYMBOL>.csv; daily closes 2000 (Bitcoin 2011) to 18 Sep 2026; daily log returns $r_t = \ln P_t - \ln P_{t-1}$ on each series' own calendar
- Cleaning rule for indices: weekdays only; a close equal to the previous close is removed (usually a filled holiday)

Setup: Run and Verify

```
# notebook sections "Data" and
# "Efficiency tests" define
# log_returns(name, start=None)
# variance_ratio(r, q) -> (VR, z, z*)
# chow_denning(r), runs_test(r), dfa_hurst(r)
sp = log_returns('sp500') # GSPC.INDX
bet = log_returns('bet') # BET
print(len(sp), sp.index[0].date(),
      sp.index[-1].date(), len(bet))
print(sp.head(3).round(5))
```

- Expected: 6 714 S&P 500 returns from 2000-01-04 to 2026-09-18; 6 670 BET returns

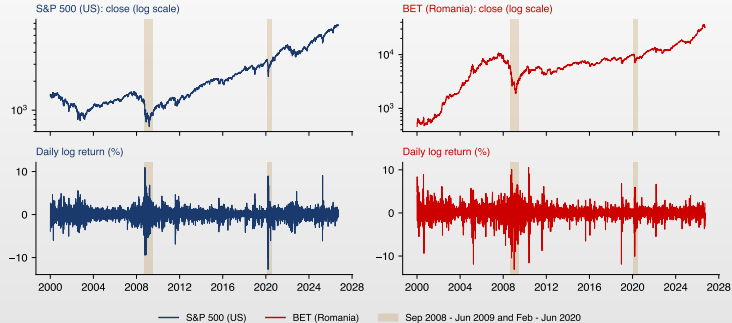
date	close	r_t
2000-01-03	1455.22	—
2000-01-04	1399.42	−0.03910
2000-01-05	1402.11	+0.00192
2000-01-06	1403.45	+0.00096

- If your count or first value differs, check the symbol, the field and the cleaning rule

Notation Used in Every Exercise

- ▣ r_t : daily log return; $e_t = r_t - \bar{r}$; T : number of returns; q : horizon in days
- ▣ $\hat{\rho}_k = \sum_{t>k} e_t e_{t-k} / \sum_t e_t^2$; robust SE $\sqrt{\hat{\tau}_k}$, $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$
- ▣ $VR(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho_k$; $z(q)$ assumes RW1 (i.i.d.), $z^*(q)$ allows volatility clustering (RW3) [Lo & MacKinlay \(1988\)](#)
- ▣ Chow–Denning $CD = \max_q |z^*(q)|$, $q = 2, 5, 10, 20$, 5% critical value 2.49 [Chow & Denning \(1993\)](#)
- ▣ Implied Hurst exponent $H(q) = 0.5 + \ln VR(q) / (2 \ln q)$, exact only for self-similar increments; DFA exponent α_{DFA} [Peng et al. \(1994\)](#)
- ▣ HAC: Newey–West errors [Newey & West \(1987\)](#); wild bootstrap: multiply e_t by random signs ± 1 [Kim \(2006\)](#); FWER: probability of at least one false rejection

The Data at a Glance



- Volatility clusters: calm years and bursts (2008–2009, 2020) alternate, so i.i.d. standard errors are too small in every B exercise

A1 [Solved]: The Variance Ratio, Line by Line — Task

- ▣ **Question:** what does $\widehat{VR}(2)$ compute, and how does it relate to $\hat{\rho}_1$?
- ▣ **Data:** eight daily returns in %, in time order: 1.0, −0.5, 0.8, 0.3, −1.2, 0.6, 0.9, −0.4; no data file; simple estimator, sums of squares divided by the number of terms
- ▣ **Tasks**
 1. Compute the mean $\hat{\mu}$, the deviations $e_t = r_t - \hat{\mu}$ and the one-day variance $\hat{\sigma}_1^2 = \frac{1}{8} \sum_t e_t^2$.
 2. Compute the lag-1 autocorrelation $\hat{\rho}_1 = \sum_{t=2}^8 e_t e_{t-1} / \sum_{t=1}^8 e_t^2$.
 3. Form the seven overlapping two-day sums $s_t = r_{t-1} + r_t$ and compute $\hat{\sigma}_2^2 = \frac{1}{7} \sum_t (s_t - 2\hat{\mu})^2$.
 4. Compute $\widehat{VR}(2) = \hat{\sigma}_2^2 / (2\hat{\sigma}_1^2)$ and compare it with $1 + \hat{\rho}_1$.
- ▣ **Report:** a worksheet with columns t , r_t , e_t , e_t^2 , $e_t e_{t-1}$, s_t , $(s_t - 2\hat{\mu})^2$ and one sentence on the sign of $\widehat{VR}(2) - 1$
- ▣ **Notebook:** section A1

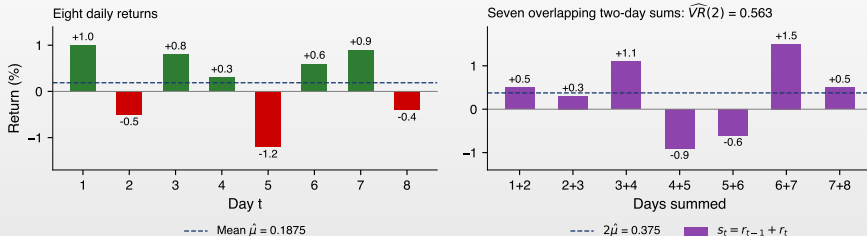
A1 [Solved]: Solution (1/2) — Worksheet

t	r_t	e_t	e_t^2	$e_t e_{t-1}$	s_t	$(s_t - 2\hat{\mu})^2$
1	+1.0	+0.8125	0.6602	—	—	—
2	−0.5	−0.6875	0.4727	−0.5586	+0.5	0.0156
3	+0.8	+0.6125	0.3752	−0.4211	+0.3	0.0056
4	+0.3	+0.1125	0.0127	+0.0689	+1.1	0.5256
5	−1.2	−1.3875	1.9252	−0.1561	−0.9	1.6256
6	+0.6	+0.4125	0.1702	−0.5723	−0.6	0.9506
7	+0.9	+0.7125	0.5077	+0.2939	+1.5	1.2656
8	−0.4	−0.5875	0.3452	−0.4186	+0.5	0.0156
sum			4.4688	−1.7639		4.4044

$$\hat{\mu} = 1.5/8 = 0.1875; \hat{\sigma}_1^2 = 4.4688/8 = 0.5586; \hat{\rho}_1 = -1.7639/4.4688 = -0.395$$

$$\hat{\sigma}_2^2 = 4.4044/7 = 0.6292; \widehat{VR}(2) = 0.6292/(2 \times 0.5586) = 0.563, \text{ against } 1 + \hat{\rho}_1 = 0.605$$

A1 [Solved]: Solution (2/2) — Chart and Interpretation

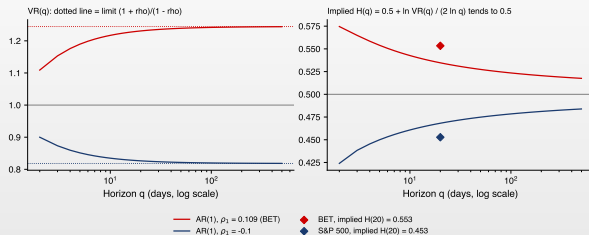


- VR(2) = $1 + \rho_1$ holds for population moments; in a sample the two differ by end terms and divisors (7 sums, 8 squares): 0.563 vs 0.605 is a convention gap, not an error
- Interpretation: consecutive returns reverse, so two-day sums vary less than two single days: mean reversion. With $T = 8$ this is arithmetic, not evidence: no test is attempted

A5 [Solved]: The Implied Hurst Exponent of BET — Task

- ▣ **Question:** does BET's $\widehat{VR}(20) > 1$ imply long memory?
- ▣ **Data:** BET, daily log returns 2000–2026 (BET, close): $\widehat{VR}(20) = 1.377$, $\hat{\rho}_1 = 0.109$ (Lecture 2); for exactly self-similar increments (fractional Gaussian noise) $VR(q) = q^{2H-1}$
- ▣ **Tasks**
 1. Take logarithms of $VR(q) = q^{2H-1}$ and solve for the implied $H(q)$.
 2. Compute the implied $H(20)$ for BET.
 3. Compute $VR(20)$ and $H(20)$ of a short-memory AR(1) with $\rho_k = 0.109^k$, using
$$VR(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho_k.$$
- ▣ **Report:** the formula, the two $H(20)$ values with three decimals, and whether one horizon can identify long memory
- ▣ Notebook: section A5

A5 [Solved]: Solution



- (1) $\ln VR(q) = (2H - 1) \ln q$, so
 $H(q) = 0.5 + \ln VR(q) / (2 \ln q)$
- (2) $\ln 1.377 = 0.3199$, $2 \ln 20 = 5.991$:
 $H(20) = 0.5 + 0.3199 / 5.991 = 0.553$
- (3) AR(1): $VR(20) = 1.231$, $H(20) = 0.535$, and
 $H(250) = 0.520$ (red curve)
- Interpretation: short memory alone gives most of
 BET's $H(20)$, and its $H(q)$ drifts back to 0.5: one
 horizon cannot identify long memory

A6 [Proposed]: The Implied Hurst Exponent of the S&P 500 — Task

- ▣ **Question:** is the S&P 500 “anti-persistent”, or only short-run reversing?
- ▣ **Data:** S&P 500, daily log returns 2000–2026 (GSPC.INDX, close): $\widehat{\text{VR}}(20) = 0.753$; a short-memory benchmark AR(1) with $\rho_k = (-0.1)^k$
- ▣ **Tasks**
 1. Compute the implied $H(20)$ of the S&P 500 with the formula of A5.
 2. Compute $\text{VR}(20)$ and $H(20)$ of the AR(1) benchmark.
 3. Compute $\text{VR}(250)$, $H(250)$ and the limit $\text{VR}(\infty) = (1 + \rho)/(1 - \rho)$ of the AR(1), and read the blue curve of the A5 chart.
- ▣ **Report:** the five numbers and one sentence: does $H(20) < 0.5$ require long memory?
- ▣ **Model:** A5 [Solved]
- ▣ Notebook: section A6

B1 [Solved]: Classical vs Robust Variance Ratio — Task

- **Question:** do the S&P 500 and BET reject the random walk once volatility clustering is allowed?
- **Data:** GSPC.INDX and BET, daily log returns 2000–2026, $T = 6\,714$ and $6\,670$; H_0 of $z(q)$: RW1 (i.i.d.); H_0 of $z^*(q)$: RW3 plus the conditions of [Lo & MacKinlay \(1988\)](#)
- **Tasks**
 1. For the S&P 500 and $q = 2$, compute $\hat{\sigma}_a^2$, $\hat{\sigma}_c^2(2)$, $\widehat{VR}(2)$, $\hat{\delta}_1$, $z(2)$ and $z^*(2)$ with the Lo–MacKinlay formulas (next slide).
 2. For both markets and $q = 2, 5, 10, 20$, compute $\widehat{VR}(q)$, $z(q)$, $z^*(q)$ and the two-sided p -values with `variance_ratio(r, q)`.
 3. Compute the Chow–Denning statistic $\max_q |z^*(q)|$ and decide at 5% with the critical value 2.49; its p -value is $1 - (2\Phi(\text{CD}) - 1)^4$.
- **Report:** a table with one row per market and q ($\widehat{VR}$, z , z^* , p), the CD decision, and one sentence on why $|z| > |z^*|$
- Notebook: section B1

B1 [Solved]: Solution (1/3) — S&P 500, $q = 2$, Step by Step

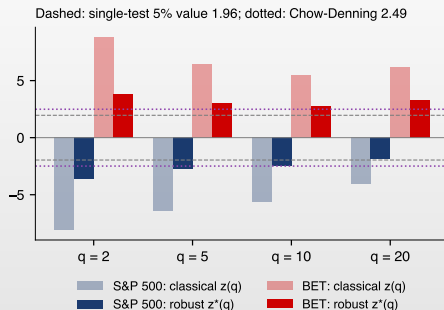
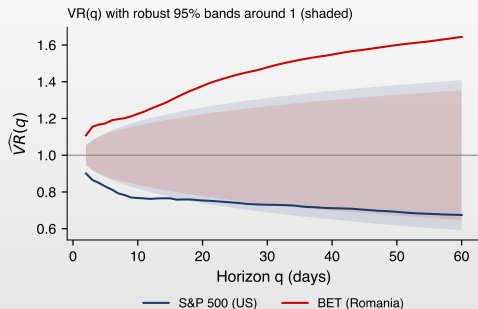
- ▣ Returns in %: $T = 6\,714$, $\hat{\mu} = 0.0247$; one-day variance $\hat{\sigma}_a^2 = \frac{1}{T-1} \sum_t e_t^2 = 1.4714$
- ▣ 6 713 overlapping sums $r_t + r_{t-1} - 2\hat{\mu}$; divisor $m = q(T - q + 1)(1 - q/T) = 13\,422$; $\hat{\sigma}_c^2(2) = 1.3256$
- ▣ $\widehat{VR}(2) = 1.3256/1.4714 = 0.9009$
- ▣ Classical: $SD = \sqrt{2(2q-1)(q-1)/(3qT)} = \sqrt{1/T} = 0.0122$, so $z(2) = (-0.0991)/0.0122 = -8.12$
- ▣ Robust: $\hat{\delta}_1 = T \sum_t e_t^2 e_{t-1}^2 / (\sum_t e_t^2)^2 = 4.966$ (it would be about 1 under i.i.d. returns); for $q = 2$, $\hat{\theta}(2) = \hat{\delta}_1$
 - ▶ $z^*(2) = \sqrt{T} (\widehat{VR}(2) - 1) / \sqrt{\hat{\theta}(2)} = 81.94 \times (-0.0991) / \sqrt{4.966} = -3.64$, $p = 0.0003$
- ▣ The robust statistic is $1/\sqrt{\hat{\delta}_1}$ times the classical one: volatility clustering makes the effective sample about 4.966 times smaller

B1 [Solved]: Solution (2/3) — Both Markets, Four Horizons

	$z(2)$	$z^*(2)$	$z(5)$	$z^*(5)$	$z(10)$	$z^*(10)$	$z(20)$	$z^*(20)$
S&P 500	-8.12	-3.64	-6.37	-2.77	-5.65	-2.50	-4.06	-1.86
BET	8.83	3.84	6.44	2.98	5.44	2.73	6.19	3.31

- ▣ $\widehat{VR}(q)$ for $q = 2, 5, 10, 20$: S&P 500 0.901, 0.830, 0.767, 0.753; BET 1.108, 1.173, 1.225, 1.377
- ▣ Chow–Denning: S&P 500 3.64 ($p = 0.0011$), BET 3.84 ($p = 0.0005$): both reject RW3 jointly

B1 [Solved]: Solution (3/3) — Chart and Interpretation



- Interpretation: reversal in the S&P 500, persistence in BET; the robust evidence is two to three times weaker, and a rejection is not a trading rule (costs, risk, stability: B7)

B7 [Proposed]: Is the S&P 500 Reversal a Crisis Effect? — Task

- ▣ **Question:** does the S&P 500 reversal of B1 survive without the two crisis periods?
- ▣ **Data**
 - ▶ GSPC.INDX, daily log returns 2000–2026, $T = 6714$; crisis days, fixed in advance: 1 Sep 2008 – 30 Jun 2009 and 15 Feb – 30 Jun 2020
 - ▶ Removing a period leaves contiguous segments. A lag pair (r_t, r_{t-1}) with $t = 1$ Jul 2009 and $t - 1 = 29$ Aug 2008 would join two days nine months apart: it is forbidden; so is any 5-day sum that spans a gap
- ▣ **Tasks**
 1. On all days compute $\hat{\rho}_1$, the i.i.d. $t = \hat{\rho}_1 \sqrt{T}$, the robust $t = \hat{\rho}_1 / \sqrt{\hat{\tau}_1}$ and $z^*(5)$.
 2. Remove the crisis days and repeat within the retained segments; report the kept returns, the valid lag pairs and the valid 5-day sums.
 3. Give robust 95% intervals for ρ_1 in both samples and decide at 5%.
- ▣ **Report:** a two-row table and one sentence: by how much does the estimate fall, and is removing crisis days legitimate?
- ▣ **Model:** B1 [Solved]
- ▣ **Notebook:** section B7

C2 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent by a student to an AI assistant: “Test whether the S&P 500 (daily closes, 2000–2026) follows a random walk with the variance ratio. Give Python code and interpret the result”

```

1  r = np.log(sp).diff().dropna()          # sp: GSPC.INDX closes, as in B1
2  T = len(r)
3  def vr(q):
4      rq = r.rolling(q).sum().dropna()    # overlapping q-day returns
5      return rq.var() / (q * r.var())
6  q = 5
7  z = (vr(q) - 1) / np.sqrt(2 * (2*q - 1) * (q - 1) / (3 * q * T))
8  p = 2 * stats.norm.sf(abs(z))

```

- **The AI's interpretation:** (a) “ $VR(2) = 0.901$, $VR(5) = 0.830$, $z(5) = -6.37$, $p < 0.001$: the random walk is rejected”; (b) “since $VR(2) = 1 + 2\rho_1$, $\rho_1 \approx -0.050$ ”; (c) “ $VR < 1$ means momentum: buy the S&P 500 after up days”

C2 [Proposed]: Critique an AI Answer — Task

- **Question:** which claims of the AI answer does its code actually support?
- **Data:** the code and interpretation on the previous slide; the same `GSPC.INDX` returns as B1
- **Tasks**
 1. Audit four points in a table (line number, claim, correct or wrong, how you check): the estimator convention, the null and its standard error, the VR identity, the economic interpretation.
 2. Say which reported numbers the code computes (it computes only $q = 5$).
 3. Give two numerical checks: $\hat{\rho}_1$ computed directly, and $z^*(5)$ from `variance_ratio`.
 4. Write the corrected interpretation in two sentences.
- **Report:** the audit table, the two checks, the corrected interpretation; log each error as in `AI_ERRORS.md`
- **Model:** A1, B1 [Solved]
- **Notebook:** section B1

B5 [Solved]: The Turn-of-the-Month Effect with HAC Errors — Task

- ▣ **Question:** do returns on turn-of-the-month days exceed those on other days?
- ▣ **Data**
 - ▶ GSPC.INDX and BET, daily log returns 2000–2026 in basis points, $r_t = 10\,000 \ln(P_t/P_{t-1})$, each on its own trading calendar; the incomplete last month (September 2026) is dropped
 - ▶ $TOM_t = 1$ on the last trading day of a month and the first three trading days of the next [Ariel \(1987\)](#), 0 otherwise; example: 31 Jul and 3, 4, 5 Aug 2026
- ▣ **Tasks**
 1. Build TOM_t from the trading calendar (rank of the day within its month, from the start and from the end).
 2. Estimate $r_t = a + b TOM_t + u_t$ by OLS; $\hat{a}$ is the mean on other days, $\hat{a} + \hat{b}$ the mean on TOM days.
 3. Compute the OLS and the Newey–West HAC standard errors of $\hat{b}$ with 5 lags [Newey & West \(1987\)](#), both t -statistics and the HAC p -value.
 4. Convert $\hat{b}$ into an approximate extra log return per month (4 TOM days), before costs.
- ▣ **Report:** a table (share of TOM days, $\hat{a}$, $\hat{b}$, both SEs and t), a 95% interval chart, one sentence on why OLS and HAC differ
- ▣ **Notebook:** section B5

B5 [Solved]: Solution (1/3) — Design Matrix

date	r_t (bp)	1	TOM _t
2026-07-27	+1.6	1	0
2026-07-28	+21.0	1	0
2026-07-29	-152.8	1	0
2026-07-30	+164.7	1	0
2026-07-31	+69.8	1	1
2026-08-03	+146.8	1	1
2026-08-04	+177.4	1	1
2026-08-05	-16.8	1	1
2026-08-06	-17.6	1	0
2026-08-07	+61.7	1	0

- S&P 500 rows around the July/August 2026 boundary: the regressors are a column of ones and the TOM dummy
- OLS: $\hat{b} = \bar{r}_{\text{TOM}} - \bar{r}_{\text{other}}$; S&P 500 $5.1 - 1.9 = 3.2$ bp; BET $20.4 - 3.1 = 17.3$ bp
- HAC: $\widehat{\text{Var}}(\hat{\beta}) = (X'X)^{-1}\hat{S}(X'X)^{-1}$, $\hat{S} = \sum_{|j| \leq 5} (1 - \frac{|j|}{6}) \sum_t x_t \hat{u}_t \hat{u}_{t-j} x'_{t-j}$: it lets the residual variance change and neighbouring days correlate
- In code: `sm.OLS(r, X).fit(cov_type='HAC', cov_kwds={'maxlags': 5})`

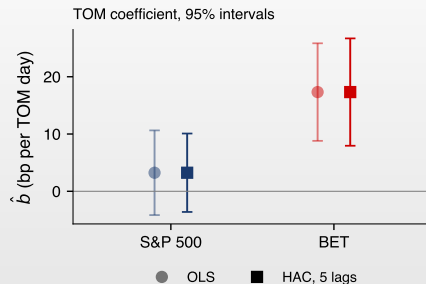
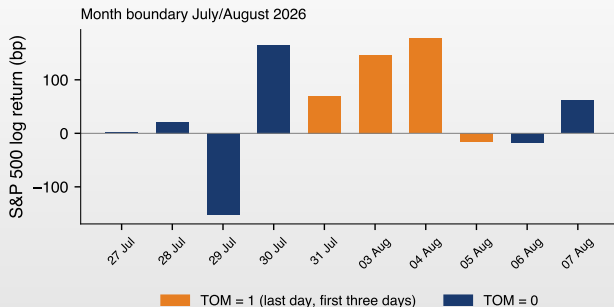
 MFM_ch2_calendar_anomalies

B5 [Solved]: Solution (2/3) — Estimates

	TOM share	$\hat{a}$ (bp)	$\hat{b}$ (bp)	SE_{OLS}	t_{OLS}	SE_{HAC}	t_{HAC} (p)
S&P 500	19.1%	1.9	3.2	3.8	0.86	3.5	0.93 (0.35)
BET	19.2%	3.1	17.3	4.3	3.99	4.8	3.62 (0.0003)

- BET: HAC 95% interval [8.0, 26.7] bp; the HAC SE exceeds the OLS SE (clustering and $\rho_1 > 0$), t falls from 3.99 to 3.62, still a rejection
- S&P 500: $[-3.6, 10.1]$ bp contains 0
- $4\hat{b} \approx 69$ bp (BET): an approximate extra monthly log return relative to the other-day mean, before costs, not a strategy return

B5 [Solved]: Solution (3/3) — Chart and Interpretation



- Interpretation: one effect in one market was picked among many: correct for the effects tried (B6) before looking for a mechanism (salary and pension inflows)

B6 [Proposed]: Fifteen Calendar Tests and Holm — Task

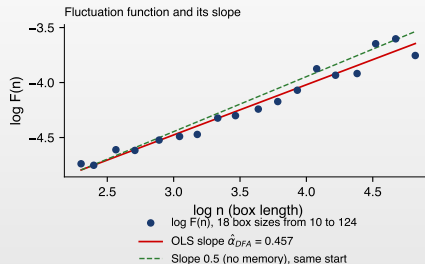
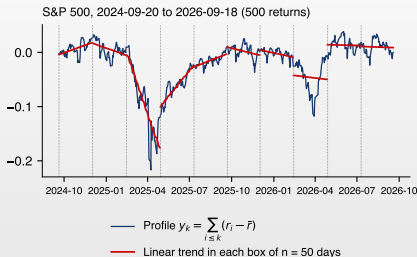
- ▣ **Question:** which of fifteen calendar effects survive a correction for multiple testing?
- ▣ **Data:** the 15 p -values below: 5 markets $\times$ 3 effects (day of week: Wald test of equal means, HAC 5 lags; January: monthly returns, HAC 3 lags; TOM: as in B5), 2000–2026
- ▣ **Tasks**
 1. Count the raw rejections at 5% and compute the FWER of 15 independent tests.
 2. Apply Bonferroni: reject if $p_i < 0.05/15$.
 3. Apply Holm: sort the p -values, compare $p_{(j)}$ with $0.05/(16 - j)$ and stop at the first failure.
- ▣ **Report:** the sorted table with ranks, thresholds and three decision columns (raw, Bonferroni, Holm), and one sentence on BUX January
- ▣ **Model:** B5 [Solved]
- ▣ **Notebook:** section B6

Market	Effect	p	Market	Effect	p
S&P 500 (US)	day of week (equal means)	0.826	Euro Stoxx 50	turn of the month	0.227
S&P 500 (US)	January	0.426	WIG20 (Poland)	day of week (equal means)	0.226
S&P 500 (US)	turn of the month	0.354	WIG20 (Poland)	January	0.686
BET (Romania)	day of week (equal means)	0.305	WIG20 (Poland)	turn of the month	0.084
BET (Romania)	January	0.137	BUX (Hungary)	day of week (equal means)	0.933
BET (Romania)	turn of the month	0.0003	BUX (Hungary)	January	0.024
Euro Stoxx 50	day of week (equal means)	0.632	BUX (Hungary)	turn of the month	0.0003
Euro Stoxx 50	January	0.774			

B3 [Solved]: Rolling DFA on the S&P 500 — Task

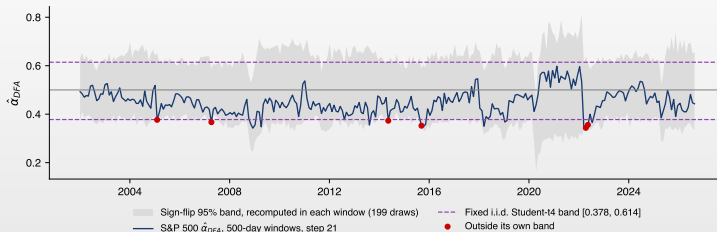
- **Question:** does the S&P 500 DFA exponent ever leave a null band that allows volatility clustering?
- **Data**
 - ▶ GSPC.INDX, daily log returns 2000–2026; windows of 500 returns moved by 21 days (296 windows); function `dfa_hurst(x)`
 - ▶ DFA grid: 18 box lengths spaced evenly in $\log n$ from 10 to 125 (a quarter of the window), rounded to integers, duplicates removed; the remainder $T - kn$ of the profile is discarded; seed 42 plus the window end
- **Tasks**
 1. For the last window (2024-09-20 to 2026-09-18) build the profile, detrend it in boxes and fit $\log F(n)$ on $\log n$.
 2. Compute α_{DFA} in every window.
 3. In each window build a sign-flip band: 199 series $e_t \eta_t$, $\eta_t = \pm 1$ with probability 1/2; keep the 2.5% and 97.5% quantiles of α_{DFA} .
 4. Compute the share of windows outside their own band and outside one fixed i.i.d. Student- t_4 band (300 simulations of length 500).
- **Report:** the one-window fit, the rolling chart, both shares, and one sentence on what the band assumes
- **Notebook:** section B3

B3 [Solved]: Solution (1/2) — One Window, Step by Step



- Profile $y_k = \sum_{i \leq k} (r_i - \bar{r})$ (left); in each box of length n fit a line and keep the residuals; $F(n)$ = root mean square of the residuals
- 18 box lengths from 10 to 124; the OLS slope of $\log F(n)$ on $\log n$ is $\hat{\alpha}_{DFA} = 0.457$ (right), close to 0.5

B3 [Solved]: Solution (2/2) — All Windows



- Outside their own band: 2.0% of 296 windows; outside the fixed Student- t_4 band [0.378, 0.614]: 6.4%
- Mean $\hat{\alpha}_{DFA} = 0.450$; minimum 0.341 (window ending 2022-03-15), maximum 0.598 (2021-02-12); the band is widest (0.500) in the window ending 2022-02-11
- Interpretation: the band is a pointwise benchmark (signs symmetric and independent given $|e_t|$), not a confidence interval; overlapping windows are correlated, so the shares describe, they do not test stability; no memory beyond chance

B4 [Proposed]: Adaptive Efficiency of Bitcoin — Task

- **Question:** has Bitcoin moved towards efficiency since its early years?
- **Data:** BTC-USD.CC, close, daily log returns 7 days a week from 1 Jan 2011 to 18 Sep 2026 (earlier than the lecture's 2014 start, to include the first bubbles); windows of 365 days moved by 30 days; H_0 : RW3 [Lo \(2004\)](#)
- **Tasks**
 1. Compute the robust $z^*(5)$ in each window.
 2. In each window build a sign-flip band for $z^*(5)$ with 999 replications (seed 42 plus the window end) and keep its 2.5% and 97.5% quantiles.
 3. Count the windows outside their band and outside ± 1.96 and compare with $0.05 \times$ the number of windows.
 4. Compare with [Urquhart \(2016\)](#): 2010–2016, a battery of classical tests (Ljung–Box, runs, Bartels, automatic VR, BDS, R/S).
- **Report:** the chart, the flagged windows with z^* and band, both counts, and one sentence: what does “no detected trend” mean?
- **Model:** B3, B1 [Solved]
- Notebook: section B4

Submission Checklist and Team Repository

Hand in before the next seminar

- ▣ A1, A5, A6: the worksheet and the $H(q)$ numbers
- ▣ B1, B7: the tables and the B1 chart
- ▣ C2: the audit table, two checks, your error log
- ▣ B5, B6: the TOM table and the sorted p -value table
- ▣ B3, B4: the rolling charts and the shares
- ▣ One interpretation sentence per exercise

Team repository

- ▣ One Quantlet folder: B1 or B3 for one market not used in class
- ▣ Log AI use in AI_USE.md and caught errors in AI_ERRORS.md
- ▣ Optional: one extension pack; C1 and C3 are possible project starts
- ▣ Rubric: data and code reproducible, computation correct, uncertainty reported, interpretation stated

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Theory	A2 (null distribution), A3 [Solved] (runs), A4 (Stambaugh bias)	A1, B1
2. Many markets	B2	B1
3. Predictability	B8	A4, B5
4. Event study	B9	Seminar 0, B7; B5
5. Project seeds	C1, C3	B1, B3; the lecture case study

- Each exercise has its own notebook section; do the packs in order, the solutions of the proposed ones are discussed in class

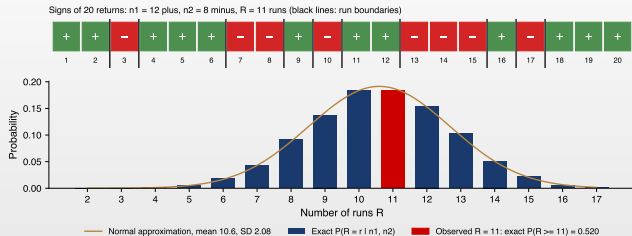
A2 [Proposed]: The Variance Ratio and Its Null Distribution — Task

- ▣ **Question:** why is $\sqrt{T}(\widehat{VR}(q) - 1)$ approximately Normal with variance $2(2q - 1)(q - 1)/(3q)$, and when does this fail?
- ▣ **Data:** r_t covariance-stationary, variance γ_0 , autocovariances $\gamma_k = \text{Cov}(r_t, r_{t-k})$, $\rho_k = \gamma_k/\gamma_0$; under RW1 $\sqrt{T}(\hat{\rho}_1, \dots, \hat{\rho}_{q-1})$ tends to $N(0, I_{q-1})$; S&P 500: $T = 6\,714$, $\widehat{VR}(2) = 0.901$
- ▣ **Tasks**
 1. For $q = 3$, write $\text{Var}(r_t + r_{t-1} + r_{t-2})$ as the sum of the nine covariances and count how many times $\gamma_0, \gamma_1, \gamma_2$ appear.
 2. Generalise: show that $\text{VR}(q) = 1 + \sum_{k=1}^{q-1} a_k \rho_k$ with $a_k = 2(1 - k/q)$.
 3. Under RW1 compute $a' I a = \sum_k a_k^2$ and simplify it to $2(2q - 1)(q - 1)/(3q)$; evaluate it for $q = 2, 5, 20$.
 4. Compute $z(2)$ for the S&P 500 and explain why I must be replaced by $\text{diag}(\delta_k)$ under RW3.
- ▣ **Report:** the counting, the variance in two columns (asymptotic, of $\sqrt{T}(\widehat{VR} - 1)$; finite-sample, of $\widehat{VR}$: divide by T) and $z(2)$
- ▣ **Model:** A1 [Solved]
- ▣ **Notebook:** section A2

A3 [Solved]: The Runs Test, Step by Step — Task

- ▣ **Question:** are 20 signs consistent with independence?
- ▣ **Data:** signs of 20 consecutive daily returns, no zeros: ++-+++--+-+---+---+; under H_0 the signs are independent with a constant probability of a positive day [Wald & Wolfowitz \(1940\)](#)
- ▣ **Tasks**
 1. Number the signs 1 to 20, mark every sign change and count n_1 , n_2 and the runs R .
 2. Compute $\mathbb{E}[R] = 2n_1n_2/n + 1$ and $\text{Var}[R] = 2n_1n_2(2n_1n_2 - n)/(n^2(n - 1))$.
 3. Compute the approximate z and its two-sided p -value.
 4. Compare with the exact conditional distribution of R given n_1 and n_2 .
- ▣ **Report:** n_1 , n_2 , R , $\mathbb{E}[R]$, $\text{Var}[R]$, z , both p -values and what too few or too many runs would mean
- ▣ **Notebook:** section A3

A3 [Solved]: Solution



- (1) Runs: ++ | - | +++ | -- | + | - | ++ | --- | + | - | +++, so $n_1 = 12$, $n_2 = 8$, $R = 11$ (10 sign changes)
- (2) $\mathbb{E}[R] = 2 \cdot 12 \cdot 8/20 + 1 = 10.60$; $\text{Var}[R] = 192 \cdot 172/(400 \cdot 19) = 4.345$; (3) approximate $z = 0.40/2.085 = 0.19$, $p = 0.85$
- (4) Exact: $P(R = 11) = 0.183$, $P(R \geq 11) = 0.520$, $P(R \leq 11) = 0.663$, so the exact two-sided p is 1. Interpretation: do not reject independence; too few runs would mean persistent signs, too many alternating signs

A4 [Proposed]: The Stambaugh Bias — Task

- **Question:** why does a persistent valuation ratio look like a significant predictor too often?
- **Data**
 - ▶ x_t : the log dividend–price ratio, very persistent; when prices rise, x falls, so the innovations of returns and of x are negatively correlated
 - ▶ **System:** $r_{t+1} = \alpha + \beta x_t + u_{t+1}$, $x_{t+1} = \theta + \rho x_t + v_{t+1}$; write $u = \phi v + e$, $\phi = \sigma_{uv} / \sigma_v^2$; Kendall: $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$ [Stambaugh \(1999\)](#)
 - ▶ Illustrative annual inputs (decimal units): $\rho = 0.98$, $T = 60$ years, $\text{corr}(u, v) = -0.9$, $\sigma_u = 0.17$, $\sigma_v = 0.14$
- **Tasks**
 1. With demeaned $\tilde{x}$, show $\hat{\beta} - \beta = \phi(\hat{\rho} - \rho) + \sum \tilde{x}_t e_{t+1} / \sum \tilde{x}_t^2$ and derive $\mathbb{E}[\hat{\beta} - \beta]$.
 2. Compute ϕ , $\mathbb{E}[\hat{\rho} - \rho]$ and the bias of $\hat{\beta}$.
 3. Compute $\text{SE}(\hat{\beta}) \approx \sigma_u / (\sigma_x \sqrt{T})$, $\sigma_x = \sigma_v / \sqrt{1 - \rho^2}$, and the bias in standard errors.
- **Report:** the derivation, ϕ , the bias, the SE, their ratio, and what a positive bias does to the false-rejection rate
- **Model:** A1 [Solved]
- **Notebook:** section A4

B2 [Proposed]: BET-TR and Emerging Markets — Task

- ▣ **Question:** are emerging markets less efficient as a group?
- ▣ **Data:** BETTR.INDX (from 2014) and WIG20.INDX, BUX.INDX, PX.INDX, XU100.INDX (BIST 100), BVSP.INDX (Bovespa), MXX.INDX (IPC), NSEI.INDX (Nifty 50), SSEC.INDX (Shanghai), close, 2000–2026, each on its own calendar
- ▣ **Tasks**
 1. Start with BET and BET-TR on the same dates (BET-TR's period): classical $z(2)$, robust $z^*(2)$, Chow–Denning.
 2. Repeat for the eight other indices and report three decisions separately: classical $q = 2$, robust $q = 2$, robust joint over four horizons.
 3. Before any group-wide claim, correct the joint p -values for the nine markets tested (Holm).
- ▣ **Report:** one table (one row per market), the matched BET comparison and one sentence: dividends or sample?
- ▣ **Model:** B1 [Solved]
- ▣ **Notebook:** section B2

B8 [Proposed]: Predictive Regression and an Out-of-Sample Test — Task (1/2)

- **Question:** does the dividend–price ratio predict S&P 500 excess returns, in and out of sample?
- **Data**
 - ▶ Monthly table 1934–2026 ($T = 1\,111$): P = Shiller S&P Composite monthly average price (to 2026-06), extended with GSPC.INDX month-end closes; D = trailing 12-month dividends (Shiller, interpolated from quarterly totals), extended with SPY distributions reconstructed from close and adjusted_close, spliced on the common period
 - ▶ $r_{t+1} = \ln((P_{t+1} + D_{t+1}/12)/P_t) - \ln(1 + \text{TB3MS}_t/1200)$ (FRED, known at t); predictor $dp_t = \ln D_t - \ln P_t$, used one month later

B8 [Proposed]: Predictive Regression and an Out-of-Sample Test — Task (2/2)

▣ Tasks

1. Estimate $r_{t+1} = \alpha + \beta dp_t + u_{t+1}$ by OLS with Newey–West errors (12 lags); estimate the AR(1) of dp and $\text{corr}(\hat{u}, \hat{v})$; correct $\hat{\beta}$ for the Stambaugh bias (A4).
2. From January 1970, at each month re-estimate on data up to the previous month and forecast; the benchmark is the historical mean of the same data; compute R_{OS}^2 [Welch & Goyal \(2008\)](#) (pseudo out-of-sample: dividends are interpolated).
3. Compute the Clark–West t (HAC, 12 lags, one-sided) [Clark & West \(2007\)](#); impose the Campbell–Thompson restrictions (slope sign, non-negative forecast) [Campbell & Thompson \(2008\)](#).

▣ **Report:** $\hat{\beta}$, t , $\hat{\rho}$, the bias, R_{OS}^2 , the Clark–West t , and why Diebold–Mariano is the wrong test for nested models

▣ **Model:** A4, B5

▣ **Notebook:** section B8

B9 [Proposed]: The Bank Tax, TLV and BRD Jointly — Task (1/2)

- ▣ **Question:** how much of the bank-tax fall survives proper event-study inference?
- ▣ **Data**
 - ▶ TLV.R0, BRD.R0 (adjusted_close) and BET (close), prices joined on common trading days, then log returns
 - ▶ Timeline: tax announced on the evening of 18 Dec 2018; day 0 = 19 Dec; estimation window days $[-250, -11]$ ($T_0 = 240$); event window $[0, +5]$
 - ▶ X_0 (240×2) and X_e (6×2): the $[1, r_{m,t}]$ rows of the estimation and event windows; ι : a 6×1 vector of ones

B9 [Proposed]: The Bank Tax, TLV and BRD Jointly — Task (2/2)

▣ Tasks

1. For TLV fit the market model on $[-250, -11]$, compute AR_0 , the naive $t = AR_0 / \hat{\sigma}_\varepsilon$ and the t with the prediction-error variance; repeat for BRD.
2. For the equal-weighted portfolio of the two banks fit the market model to its own returns and compute its AR_0 and t ; compare with the t that assumes independent banks.
3. Compute $CAR[0, +5]$ and $CAR[+1, +5]$ with $\text{Var}(CAR) = \iota' \hat{\sigma}_\varepsilon^2 [\mathbf{I}_6 + X_e(X_0' X_0)^{-1} X_e'] \iota$ (adapted to the window used).
4. Compute the residual correlation $\bar{r}$ and the Kolari–Pynnönen factor $\sqrt{(1 - \bar{r}) / (1 + (N - 1)\bar{r})}$ [Kolari & Pynnönen \(2010\)](#), a correction for cross-sectional tests with many firms.

▣ **Report:** one table (TLV, BRD, portfolio), the AR/CAR chart, and whether the reversal contradicts semi-strong efficiency; note that BET contains both banks

▣ **Model:** Seminar 0, B7; B5

▣ **Notebook:** section B9

C1 [Proposed]: Did the BVB Become More Efficient after the 2020 FTSE Upgrade? — Task (1/2)

- ▣ **Question:** did BET's lag-1 autocorrelation move towards zero after the upgrade, relative to WIG20 and BUX?
- ▣ **Data**
 - ▶ [FTSE Russell \(2019\)](#): announced 26 Sep 2019, effective 21 Sep 2020 (Frontier to Secondary Emerging); background: [Pele & Mazurencu-Marinescu \(2012\)](#)
 - ▶ BET, WIG20.INDX, BUX.INDX, close; before: 21 Sep 2015 – 20 Sep 2020; after: 21 Sep 2020 – 21 Sep 2025; later data only as a visibly separated extension

C1 [Proposed]: Did the BVB Become More Efficient after the 2020 FTSE Upgrade? — Task (2/2)

▣ Tasks

1. For each market compute $\hat{\rho}_1$ with robust SE and $z^*(5)$ in both windows.
2. Build a 95% interval for the change in $\hat{\rho}_1$ by a moving-block bootstrap, independent within each period (blocks of 20 days, 999 replications, seed 42) [Künsch \(1989\)](#).
3. Plot rolling 250-day $\hat{\rho}_1$ with robust intervals for the three markets, marking the announcement, the effective date and the end of the formal window.
4. List the confounders (COVID-19, the 2023 Hidroelectrica listing, liquidity) and say what a control-adjusted contrast would need.

▣ **Report:** a Quantlet folder and one slide: conclusion, main limitation, link with the AMH [Lo \(2004\)](#); judge movement towards zero, not the sign

▣ **Model:** B1, B3 [Solved]

▣ **Notebook:** section C1

C3 [Proposed]: Big Data and Efficiency, In and Out of Sample — Task (1/2)

- **Question:** with 238 predictors, is in-sample predictability of industry returns matched out of sample?
- **Data**
 - ▶ Lecture 2 case study [Martin & Nagel \(2022\)](#): rational learning makes in-sample tests reject although out-of-sample returns are unpredictable; replicate their Sec. 7 design
 - ▶ 49 industry portfolios, [Kenneth French Data Library](#), value-weighted monthly returns, July 1926 to the last month; missing values −99.99 and −999 dropped month by month; returns divided by 100
 - ▶ Dimensions: in month t , X_t is $N_t \times 238$ (returns and squared returns at lags 2–120), N_t the industries with a complete 120-month history; h_t the 238×1 coefficients

C3 [Proposed]: Big Data and Efficiency, In and Out of Sample — Task (2/2)

▣ Tasks

1. Demean the returns and each predictor across industries every month, scale each predictor to unit cross-sectional SD, give each month equal weight.
2. Estimate pooled OLS h_t over the 240 months ending in t (data up to t only); compute $roos_{t+1} = \frac{1}{N} h_t' X_{t+1}' r_{t+1}$ and rs over the same window; state the first forecast month.
3. Average $roos$ over 10-year windows with descriptive ± 2 SE bands (Fig. 8).
4. Fit a ridge from January 1971 with standardised predictors; choose the penalty on a log grid by leave-one-year-out CV (it may use later years: a validation, not a forecast); plot the coefficients by lag.

▣ **Report:** the chart of step 3, early and late means of rs and $roos$ with SE, the coefficient plot, and why 49 portfolios can differ from thousands of stocks

▣ **Model:** B5, B8

▣ **Notebook:** section C3 (lecture Quantlet MFM_ch2_big_data)

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